Ilja Safronovs

R&D engineer, 3D reconstruction. Giraffe360, Riga, Latvia.

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I work on 3D Gaussian splatting at Giraffe360: training from sparse panoramas, in PyTorch and CUDA. Most of my work starts with a reconstruction that looks wrong. **I find the cause and fix it in our training code.** I mentored a junior engineer and led three engineers on a new 3D reconstruction feature. **Next, I want to grow into leading a team.**

Experience

Giraffe360, R&D engineer — Riga, 2025 to present

Giraffe360 makes cameras and software for real-estate virtual tours. I am in the R&D team, on the 3D reconstruction pipeline.

- **Blurry windows, outdoor views and mirrors in our 3D videos.** I found the cause and tested fixes in training runs until the haze and floaters were gone. Builds on the team's earlier window masks. **Default for PRO camera projects since June 2026.** Before and after: instagram.com/p/DZsNRB8o772
- **Our training defaults use my Gaussian pruning and relocation rules** (Jan to Jul 2026). They build on Taming 3DGS (Mallick et al., 2024) and the team's influence tracking: a Gaussian that adds little to the renders gets pruned or moved.
- **Implemented Analytic-Splatting (Liang et al., 2024) in gsplat**, CUDA forward and backward passes. We use it for downsampling.
- Cut GPU idle time in training.
- **Looped drone video**, one flight per room. My idea from the Feb 2026 hackathon (3 of my ideas were recognised out of 51). I wrote the camera path and reworked the video pipeline. The look was worked out with the designers. **It is a product now**: instagram.com/giraffe360/reel/Dd9ODBVIHhE
- **Exterior orbital videos** (Oct 2026): the toolkit that orients, selects and crops the outdoor scene, and the per-frame video pipeline.
- Fixed the camera height in rooms on a half floor, for all video types. A long-standing bug.
- **Led three engineers on a new 3D reconstruction feature** (Oct 2026). Most of the code is mine; I reviewed and merged theirs and cut the release. In production. I have cut the releases of our 3D pipeline since September 2026.
- **Mentored a junior engineer on a new depth method** (denser LiDAR depth). Mostly his work. Our splat training has run on it since October 2026.
- **Team tools.** An experiment runner: an AI agent starts and tracks trainings on a local GPU. An MCP server (Node.js) for our task board. Ground-truth datasets and checks for splat quality. I also led the move of our 3DGS code into one monorepo, with a code-size check (2026).
- **A Rust/wgpu splat viewer that other apps can embed and control** (May 2026). It swaps and combines PLY scenes live, and shows point clouds. Before that, a side-by-side comparison tool with camera sync.
- **Prototypes**: mirror reflections done right, meshes from splats, a stage that cuts the splat count for viewers, a fly-in video intro. Got Difix and NVIDIA's fixer running on our GPUs for the team to try (Feb 2026).
- Jul to Oct 2025: a small service that wrote tour narration with the OpenAI API and AWS S3.

Education

Riga Technical University, Professional Bachelor, Financial Engineering — 2024 to 2028 (expected)

Probability, statistics, stochastic analysis, Monte Carlo methods, algorithms, databases.

AI Talent Program (Ponder) — Nov 2024 to Jan 2025

C++, Python, maths and AI, with mentors from industry. That is how I got the Giraffe360 job.

Riga State Classical Gymnasium, programming and mathematics track

State mathematics olympiads. Small Mathematics University (University of Latvia).

Projects

- **Portfolio site** (2025 to 2026): a splat of a real flat, trained with gsplat MCMC on the public Zip-NeRF "london" scene (Barron et al., 2023, CC BY 4.0). 30k steps, about 18 minutes on one RTX 3090, 28.3 dB PSNR on held-out views. A 1 MB preview loads first, then the 350k-splat model. three.js and Spark, no build step.
- **sort_visualization** (2022, C++/SFML): 16 sorting algorithms animated as columns, points, circles, spirals and pyramids. The same C++ runs in the browser, compiled to WebAssembly.
- **monte-carlo-finance** (2026, Python): option pricing with variance reduction, GARCH volatility, VaR backtests with the Kupiec test, efficient frontier, CAPM. From RTU coursework.
- **load-balancing-experiments** (2026, Python), for an RTU networks course: DNS round robin and anycast measured from Riga, a queueing simulation of balancing policies, consistent, rendezvous and Maglev hashing, a small load balancer with health checks, packet capture.
- **crypto-quant-lab** (2026, Python): backtests of drift and volatility-targeting rules on 6 coins with a 1% fee.
- **flow** (2026, Dart/Flutter, private): offline workout tracker. Live rest timers, volume and load (ACWR) analysis, JSON/CSV/PDF export, CI with golden-image tests.
- **Coding-agent harness** (private, own time): one master session plans and checks the work; workers run in their own git worktrees.
- **projects_with_NEAT** (2025, Python): NEAT agents that learn Flappy Bird and Pong.
- **RTU_program_parser** (2025, Python, with Aleksejs Peņkovs): scrapes the RTU timetable and finds extra lectures that fit your schedule.
- **Offline Graphing Calculator** (2023, with Iļja Volinskis): 4th of 27 entries in the Riga Business School programming competition for school students.

Skills

Python, C++, CUDA, PyTorch, gsplat, Rust/wgpu, Dart/Flutter, JavaScript/three.js, Java, SQL.
Linux, Docker, Git and GitLab CI, AWS S3, Node.js (MCP servers), OpenAI API.
Latvian, English, Russian.

With the R&D team at Giraffe360. Methods build on Taming 3DGS (Mallick et al., 2024) and Analytic-Splatting (Liang et al., 2024).